KPR Institute of Engineering and Technology
(Autonomous)
Avinashi Road, Arasur, Coimbatore - 641 407

Department of Electronics and Communication Engineering

AUTOMATIC PLANT WATERING SYSTEM WITHOUT MICROCONTROLLER

INNOVATIVE / APPLICATION ORIENTED

PROJECT REPORT

Submitted by

MATHUMITHA M	**25EC124**
IRFANA BARVEEN A	**25EC079**
MALINI K	**25EC121**
HARESH S	**25EC066**

In partial fulfilment for the completion of

U25EC304 & Analog Electronics Laboratory

During the academic year

2026 - 2027 (Odd Semester)

YEAR: II YEAR

TABLE OF CONTENTS

S.No.	TITLE	PAGE No.
1	Abstract	3
2	Introduction	4
3	Design	7
4	Implementation	19
5	Result and Discussion	25
6	References	30

ASSESSMENT

PERFORMANCE INDICATOR	MARKS
Creative Thinking / Novelty	/ 10
Execution	/ 10
Result	/ 20
TOTAL	/ 40

LAB IN-CHARGE

1. ABSTRACT

Watering plants regularly is one of the most common problems in plant care, especially when the owner is away from home for several days. Over-watering damages the roots, while under-watering dries the plant. Most automatic watering solutions use a microcontroller such as an Arduino, which adds cost, wiring and programming effort. This project presents an **Automatic Plant Watering System** that is built entirely from basic analog components and does not need any microcontroller or program code.

A soil moisture sensor continuously monitors the soil. Its output controls a chain of three BC547 NPN transistors, which switch a small 5 V DC water pump. The first transistor (Q1) discharges a timing capacitor whenever the soil is dry. The second transistor (Q2) works as an inverter, and the third transistor (Q3) drives the pump. A resistor–capacitor (RC) timer, made of a 100 kΩ potentiometer and a 680 µF electrolytic capacitor, keeps the pump running for an adjustable time after the soil becomes wet, so that the soil is watered properly and the pump does not switch ON and OFF rapidly. A 1N4007 diode connected across the pump protects the driver transistor from the voltage spike produced when the motor is switched off.

The circuit was designed using standard component values, verified by calculation, simulated in TinkerCAD and assembled on a breadboard with a potted plant, a water bottle and a mini submersible pump. It works from a single 5 V supply, the watering time can be adjusted with the potentiometer, and the whole circuit costs only a few hundred rupees. This report describes the theory of each part of the circuit, the design calculations, the step-by-step implementation and testing procedure, the results, the limitations and the possible improvements of the system.

Keywords: Soil moisture sensor, BC547 transistor, RC timer, inverter, flyback diode, water pump, automatic irrigation.

2. INTRODUCTION

2.1 Background

Plants are an important part of homes, offices and cities. They improve air quality, reduce stress and make a place look better, and growing plants at home has become very popular. However, a plant stays healthy only if it gets the right amount of water at the right time. Each plant has its own water requirement, and the requirement also changes with the temperature, the sunlight and the size of the pot.

In large-scale agriculture, irrigation is already automated with timers, sensors and controllers. For small potted plants, however, watering is still done by hand. People forget to water, travel for a few days, or water too much because they are not sure whether the soil is really dry. The result is a dried or rotted plant. An automatic watering system that checks the soil and supplies water only when it is needed solves this problem.

2.2 Problem Statement

To design and build a low-cost system that senses the moisture of the soil in a flowerpot and switches a small water pump ON when the soil is dry and OFF after sufficient watering, without using a microcontroller, so that the system is simple, easy to understand, easy to repair and can be built from components that are available in any electronics shop.

2.3 Why a Microcontroller-Free Design?

Many watering projects are built with an Arduino or similar board. These are flexible, but for a single potted plant they are more than what is needed. A microcontroller-free design has several advantages, especially for a laboratory project in Analog and Digital Electronics:

- **Lower cost:** only three general-purpose transistors, a few resistors, a capacitor and a diode are required in addition to the sensor and the pump.
- **No programming:** there is no code to write, upload or debug, so anyone can build the circuit by following the diagram.
- **Easy troubleshooting:** every stage can be checked by measuring a voltage with a multimeter.
- **Learning value:** the project uses transistor switching, an inverter and an RC time delay, which are the basic building blocks taught in the laboratory course.

- **Reliability:** a circuit with no firmware cannot hang or restart, and it needs very little power.

2.4 Objectives

- To sense the moisture level of the soil using a soil moisture sensor.
- To switch a mini DC water pump automatically using BC547 transistors, without a microcontroller.
- To provide an adjustable watering time using an RC timer, so that the pump does not switch ON and OFF rapidly near the dryness threshold.
- To protect the driver transistor from the inductive voltage spike of the pump motor.
- To simulate, assemble and test the circuit, and to verify its operation with a real potted plant.
- To keep the design low in cost, simple, and powered from a single 5 V supply.

2.5 Scope of the Project

The system is intended for indoor potted plants, balcony plants, small kitchen gardens and for watering plants during a short vacation. One circuit controls one pot. For several pots with different water needs, the same circuit can be repeated for each pot using a common 5 V supply. The project does not include remote monitoring, data logging or a display, since the aim is to achieve automatic watering with the simplest possible electronics.

2.6 Comparison of Watering Methods

Table 1 compares the proposed system with other common ways of watering a plant. The proposed circuit responds to the real condition of the soil like a microcontroller-based system, but without programming and at a lower cost. The reference article estimates the cost of a transistor-based system at roughly $8–15, compared with $25–35 for Arduino-based systems [1].

Table 1: Comparison of different watering methods

Method	Responds to soil moisture	Programming needed	Relative cost	Best suited for
Manual watering	No (depends on the person)	No	Very low	Attentive owners, short absences
Timer-based watering	No (fixed schedule)	No	Low to medium	Gardens with a regular water need
Microcontroller-based system	Yes	Yes	Medium to high	Feature-rich and IoT systems
Proposed transistor-based system	Yes	No	Low	Potted plants and learning projects

2.7 Applications

- Watering indoor plants and balcony pots automatically.
- Keeping plants alive while the owner is on vacation.
- Small kitchen gardens, nurseries and terrace gardens (one circuit per pot).
- Classroom demonstration of transistor switching, inverters and RC timers.
- A starting point for more advanced irrigation projects.

2.8 Organisation of the Report

The remaining part of the report is organised as follows. Section 3 (Design) explains the block diagram, each component, the circuit diagram, the working principle and the design calculations. Section 4 (Implementation) describes the TinkerCAD simulation, the hardware assembly, the connection list, the power supply, the calibration and testing procedure, the precautions and the cost. Section 5 (Result and Discussion) presents the observations, the voltage levels, the effect of the potentiometer, the advantages, the limitations, the troubleshooting guide, the conclusion and the future scope. Section 6 lists the references.

3. DESIGN

3.1 Overview and Block Diagram

The system is divided into functional blocks, shown in Fig. 1. The blocks work in three stages: **detection**, **timing** and **pump control**.

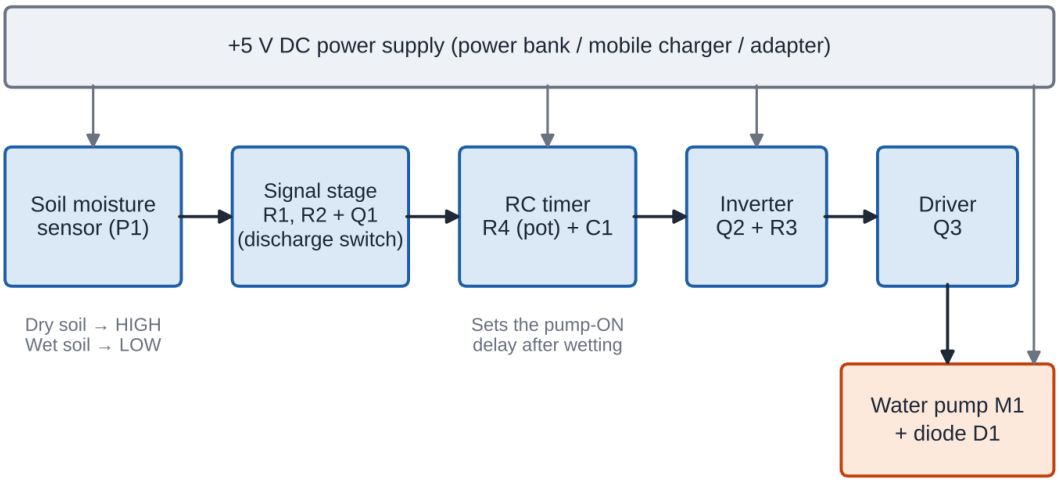


Fig. 1: Block diagram of the automatic plant watering system

- **Soil moisture sensor (P1):** measures the moisture of the soil. For the module used, the signal output is HIGH when the soil is dry and LOW when the soil is wet.
- **Signal stage (R1, R2, Q1):** R1 and R2 feed the sensor signal to the base of Q1. When the signal is HIGH, Q1 conducts and discharges the timing capacitor.
- **RC timer (R4, C1):** the potentiometer R4 and the capacitor C1 produce an adjustable time delay. When Q1 is OFF the capacitor charges slowly through R4.
- **Inverter (Q2, R3):** Q2 changes the timer voltage into the opposite logic level, so that the pump driver is ON when the soil is dry.
- **Driver (Q3):** Q3 is a switch that connects the pump to ground. The pump runs when Q3 conducts.
- **Pump (M1) and diode (D1):** the DC pump pushes water to the plant. The diode D1 protects Q3 from the voltage spike when the motor is turned OFF.
- **Power supply:** a single regulated 5 V DC source powers every block.

3.2 Description of the Components

3.2.1 Soil Moisture Sensor

A resistive soil moisture sensor has two exposed metal probes that are pushed into the soil. Dry soil is a poor conductor, so the resistance between the probes is high. When the soil contains water and dissolved salts, it conducts better and the resistance between the probes falls. The sensor board converts this change of resistance into a voltage. Many such modules also contain a comparator (for example an LM393) and a small trimmer potentiometer, which sets the moisture level at which the digital output changes between HIGH and LOW. The module used in this project has three pins:

- **VCC:** supply input, 3.3 V to 5 V (connected to +5 V).
- **GND:** ground.
- **SIG:** signal output, which is HIGH when the soil is dry and LOW when the soil is wet.

The probes are in direct contact with wet soil, so they slowly corrode, and the sensor should be checked and cleaned from time to time. Capacitive sensors do not have this problem and can be used for long-term installations.

3.2.2 BC547 NPN Transistor

The BC547 is a small-signal NPN bipolar junction transistor (BJT) in a TO-92 plastic package. It has three terminals: collector (C), base (B) and emitter (E). A small current into the base controls a larger current from the collector to the emitter. In this project all three transistors (Q1, Q2 and Q3) are BC547 and are used only as ON/OFF switches. The pin arrangement is shown in Fig. 2 and the main ratings are given in Table 2.

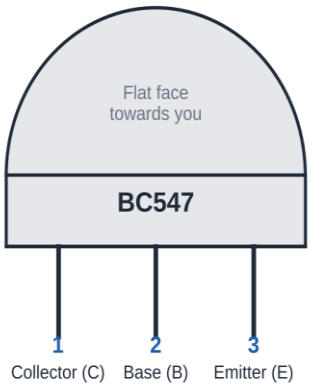


Fig. 2: Pin arrangement of the BC547 (flat face towards the reader, leads pointing down)

Table 2: Main ratings of the BC547 transistor

Parameter	Symbol	Value
Collector–emitter voltage	VCEO	45 V
Collector–base voltage	VCBO	50 V
Emitter–base voltage	VEBO	6 V
Collector current (continuous)	IC	100 mA
Power dissipation	PD	about 500 mW
DC current gain	hFE	110 to 800 (depends on the gain group)
Base–emitter voltage when ON	VBE(on)	about 0.6 V to 0.7 V
Collector–emitter saturation voltage	VCE(sat)	up to about 0.25 V
Package	—	TO-92

A transistor has three regions of operation. In the **cut-off** region the base–emitter junction is not forward biased (VBE below about 0.6 V), no collector current flows and the switch is OFF. In the **saturation** region the base is driven hard, VCE falls to about 0.1–0.2 V and the switch is fully ON. Between them is the **active** region, where the transistor works as an amplifier. In a switching circuit the transistor is made to move quickly between cut-off and saturation.

3.2.3 Resistors

Three fixed resistors are used. Their values and functions are given in Table 3. The 500 Ω value of R1 is not a standard preferred value; a 470 Ω or 510 Ω resistor can be used in its place without any noticeable change.

Table 3: Resistors used in the circuit

Ref.	Value	Colour bands (4-band)	Function
R1	500 Ω	Green, black, brown, gold	Limits the base current of Q1 from the sensor output
R2	10 kΩ	Brown, black, orange, gold	Pull-down: keeps Q1 firmly OFF when the sensor output is LOW
R3	1 kΩ	Brown, black, red, gold	Collector load of Q2 and base current source of Q3

3.2.4 Potentiometer (R4)

R4 is a 100 kΩ linear potentiometer. It is used as a variable resistor between the +5 V rail and the timer node. Turning the knob changes the charging resistance of the RC timer and therefore the time for which the pump keeps running after the soil becomes

wet. A larger resistance gives a longer delay and a smaller resistance gives a shorter delay.

3.2.5 Electrolytic Capacitor (C1)

C1 is a 680 µF polarised electrolytic capacitor. It stores the timing charge. Its negative lead is connected to ground and its positive lead to the timer node. The voltage across C1 never exceeds 5 V in this circuit, so a capacitor with a rating of 10 V or more (16 V is common) is suitable. Connecting an electrolytic capacitor with reversed polarity can damage it, so the marking on its body must be checked. The value of C1 can be changed by trial to suit the plant, because the delay is proportional to the capacitance.

3.2.6 Diode 1N4007 (D1)

The 1N4007 is a general-purpose silicon rectifier diode with a reverse voltage rating of 1000 V and an average forward current of 1 A. Here it is used as a **flyback (freewheeling) diode** across the pump. It is connected with its cathode to +5 V and its anode to the collector of Q3, so that it does not conduct while the pump is running and conducts only for a short time when the motor current is interrupted.

3.2.7 Mini DC Water Pump (M1)

The pump is a small submersible DC motor pump rated for 3 V to 6 V. Inside there is a brushed DC motor that rotates an impeller. When the pump is placed in the water container and connected to a tube, it pushes water to the plant. The pump must be waterproofed at the wire joints, and it must never run dry for a long time. Since the motor is an inductive load, it needs the flyback diode D1.

3.2.8 Power Supply and Breadboard

The circuit is designed for a 5 V DC supply. This can be a power bank through a USB breakout board, a mobile phone charger or a regulated 5 V adapter. The supply should be able to deliver at least 500 mA, because the pump draws a much larger current than the electronics. A solderless breadboard is used to assemble the circuit, because components can be changed easily.

3.3 Bill of Materials

Table 4: List of components used

S.No.	Component	Value / Type	Quantity
1	Soil moisture sensor module (P1)	Resistive, 3-pin	1
2	NPN transistor (Q1, Q2, Q3)	BC547	3
3	Electrolytic capacitor (C1)	680 µF, 10 V or more	1
4	Resistor (R1)	500 Ω, ¼ W	1
5	Resistor (R2)	10 kΩ, ¼ W	1
6	Resistor (R3)	1 kΩ, ¼ W	1
7	Potentiometer (R4)	100 kΩ, linear	1
8	Diode (D1)	1N4007	1
9	Mini submersible water pump (M1)	DC 3–6 V	1
10	Breadboard	Standard size	1
11	Connecting wires, water tube	As required	—
12	5 V DC power source	Power bank / adapter	1

3.4 Circuit Diagram

The complete circuit is shown in Fig. 3. The sensor P1 is powered from +5 V. Its signal output goes through R1 to the base of Q1, and R2 connects the base of Q1 to ground. The collector of Q1 is connected to the **timer node**, where the wiper of R4, the positive lead of C1 and the base of Q2 meet. The emitters of Q1, Q2 and Q3 are connected to ground. The collector of Q2 is connected to +5 V through R3 and directly to the base of Q3. The pump M1 is connected between +5 V and the collector of Q3, and the diode D1 is connected across the pump.

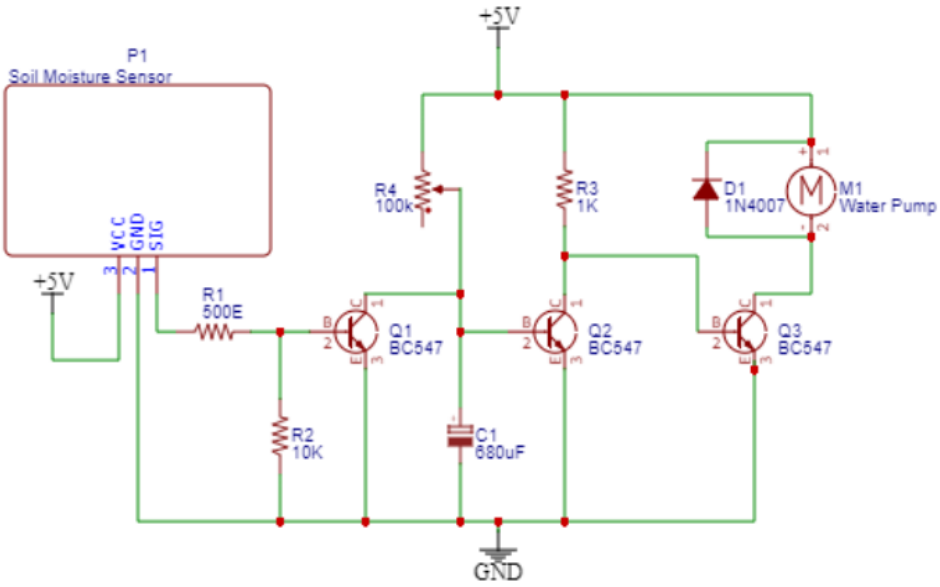


Fig. 3: Circuit diagram of the automatic plant watering system

3.5 Theory of the Circuit Blocks

3.5.1 Transistor as a Switch

When the base of an NPN transistor is driven with enough current, the transistor saturates and the collector is pulled almost to ground, like a closed switch. With no base current the transistor is cut off and behaves like an open switch. To guarantee saturation, the base current is chosen to be several times larger than the minimum value Ic / hFE. This is called overdriving, and a forced gain of about 10 is a common design rule.

$$I_{B(min)} = I_C / h_{FE} \qquad \text{and} \qquad I_{B(design)} \approx I_C / 10$$

3.5.2 Transistor as an Inverter

Q2 has a resistor R3 between its collector and +5 V. When the base of Q2 is low, Q2 is OFF and the collector is pulled HIGH by R3. When the base of Q2 is high, Q2 is ON and the collector is pulled LOW. The output is therefore the opposite of the input, and Q2 acts as a NOT gate. The inverter is needed because most soil moisture modules give a HIGH output for dry soil, while the pump driver must be ON in that condition. A sensor that gives LOW for dry soil would not need Q2.

3.5.3 RC Timer

A capacitor C charged from a supply voltage VCC through a resistor R rises according to the equation

$$Vc(t) = VCC\ (1 - e^{(-t / RC)})$$

where RC is the **time constant** in seconds. Q2 starts conducting when the voltage at its base reaches about VBE ≈ 0.7 V. Putting Vc = 0.7 V and VCC = 5 V in the equation and solving for t gives the delay:

$$t = -RC \cdot \ln(1 - 0.7 / 5) \approx 0.151 \times R \times C$$

The delay is directly proportional to both R and C. This is why the potentiometer, which changes R, can set the watering time, and why a larger C1 gives a longer pump-ON time.

3.5.4 Inductive Load and the Flyback Diode

A motor winding is an inductor. The current in an inductor cannot change instantly, so when Q3 suddenly interrupts the motor current, the inductor forces the current to continue and the voltage at the collector of Q3 rises sharply (v = L di/dt). Without protection, this spike can exceed the 45 V rating of the transistor and damage it. The diode D1 gives this current a path back to the supply. The collector voltage is then clamped to about VCC + 0.7 V, which is about 5.7 V, and the transistor is safe.

3.6 Working of the Circuit

The operation can be understood in two situations.

Case 1: the soil is dry. The sensor output is HIGH. Current flows through R1 into the base of Q1, so Q1 saturates and holds the timer node near 0 V. C1 is therefore discharged. With the base of Q2 at about 0 V, Q2 is OFF. The resistor R3 now supplies base current to Q3, so Q3 is ON and the pump runs and waters the plant.

Case 2: the soil becomes wet. The sensor output goes LOW and Q1 turns OFF. The timer node is no longer held low, and C1 begins to charge from +5 V through R4. While the voltage across C1 is below about 0.7 V, Q2 stays OFF and Q3 stays ON, so the pump keeps running and the water spreads through the soil. When the voltage reaches about 0.7 V, Q2 turns ON and pulls the base of Q3 to ground. Q3 turns OFF and the pump stops. The pump stays OFF until the soil becomes dry again, when Q1 turns ON, discharges C1 and the cycle starts again.

The delay between the soil becoming wet and the pump stopping is set by R4 and C1. This delay gives the water time to soak into the soil and prevents the pump from switching ON and OFF repeatedly when the soil moisture is close to the sensor threshold. The states of all the parts are summarised in Table 5, the operation is shown

as a flowchart in Fig. 4 and the voltages and the pump state are drawn against time in Fig. 5.

Table 5: Operating states of the circuit

Soil condition	Sensor output	Q1	Q2	Q3	Pump
Dry	HIGH	ON	OFF	ON	ON
Wet, C1 charging (below 0.7 V)	LOW	OFF	OFF	ON	ON (delay period)
Wet, C1 charged (about 0.7 V)	LOW	OFF	ON	OFF	OFF

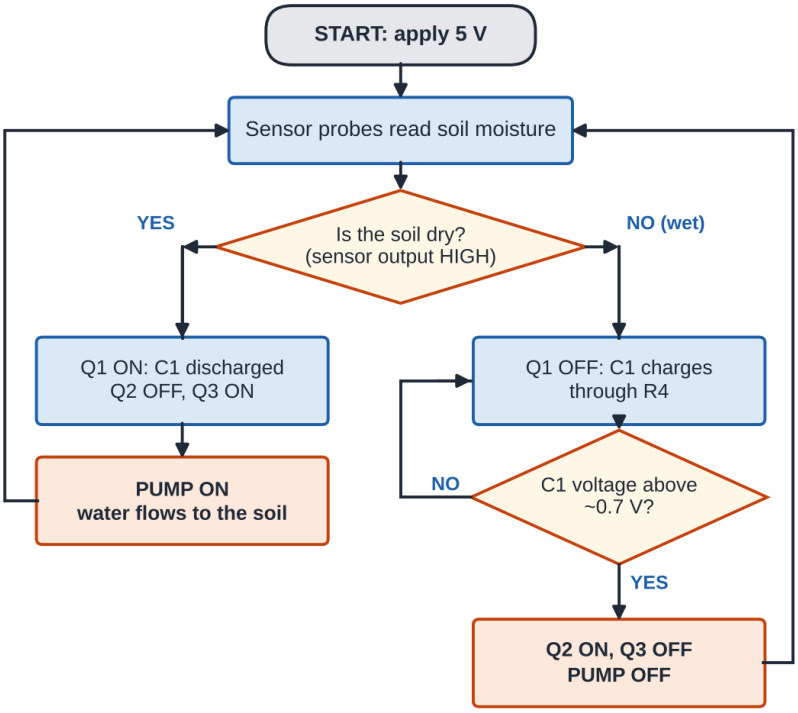


Fig. 4: Flowchart of the operation of the system

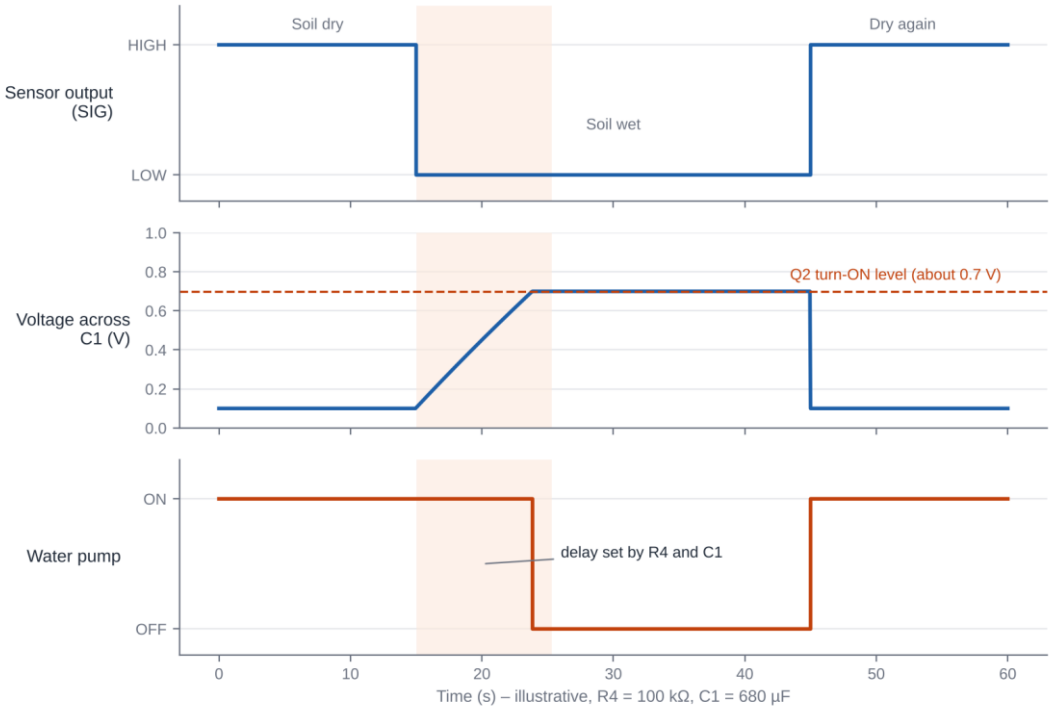


Fig. 5: Timing diagram: sensor output, voltage across C1 and pump state (illustrative)

3.7 Design Calculations

3.7.1 Base Drive of Q1 (R1 and R2)

With a sensor output of 5 V (HIGH) and VBE(Q1) = 0.7 V, the current through R1 is (5 − 0.7) / 500 = 8.6 mA. R2 takes 0.7 V / 10 kΩ = 0.07 mA, so the current entering the base of Q1 is about

$$IB(Q1) = 8.6\ mA - 0.07\ mA \approx 8.5\ mA$$

The collector current of Q1 is only the small current from R4 into the timer node, which is at most (5 − 0.2) V / 100 kΩ ≈ 48 µA. The base current needed for hard saturation with a forced gain of 10 is only about 5 µA. The base drive is therefore much larger than required, and Q1 saturates reliably even if the sensor output can supply only a part of the 8.5 mA. This is important because some modules have a weak output (an LM393 output with a pull-up resistor). R2 ensures that the base of Q1 is pulled to ground when the sensor output is LOW, so Q1 does not turn ON because of leakage or noise.

3.7.2 Minimum Resistance of R4

When Q1 is ON, the current through R4 flows into the collector of Q1. If R4 is turned to a very small value, this current becomes large. At 1 kΩ it is about 4.8 mA, which is still safe, but at 0 Ω nothing limits it except the transistor itself and Q1 could be

damaged. For this reason the potentiometer should not be set to zero resistance, or a fixed resistor of about 1 kΩ should be connected in series with R4 as protection.

3.7.3 Delay of the RC Timer

Using t = 0.151 × R4 × C1, the pump-ON delay for different potentiometer settings and capacitor values is calculated in Table 6. The charging curves for C1 = 680 µF are plotted in Fig. 6. For the 100 kΩ and 680 µF combination the time constant is 68 s and the delay is about 10 s.

Table 6: Calculated pump-ON delay for different R4 and C1 values

R4 in the circuit	C1 = 470 µF	C1 = 680 µF	C1 = 1000 µF
100 kΩ	7.1 s	10.3 s	15.1 s
75 kΩ	5.3 s	7.7 s	11.3 s
50 kΩ	3.5 s	5.1 s	7.5 s
25 kΩ	1.8 s	2.6 s	3.8 s
10 kΩ	0.7 s	1.0 s	1.5 s

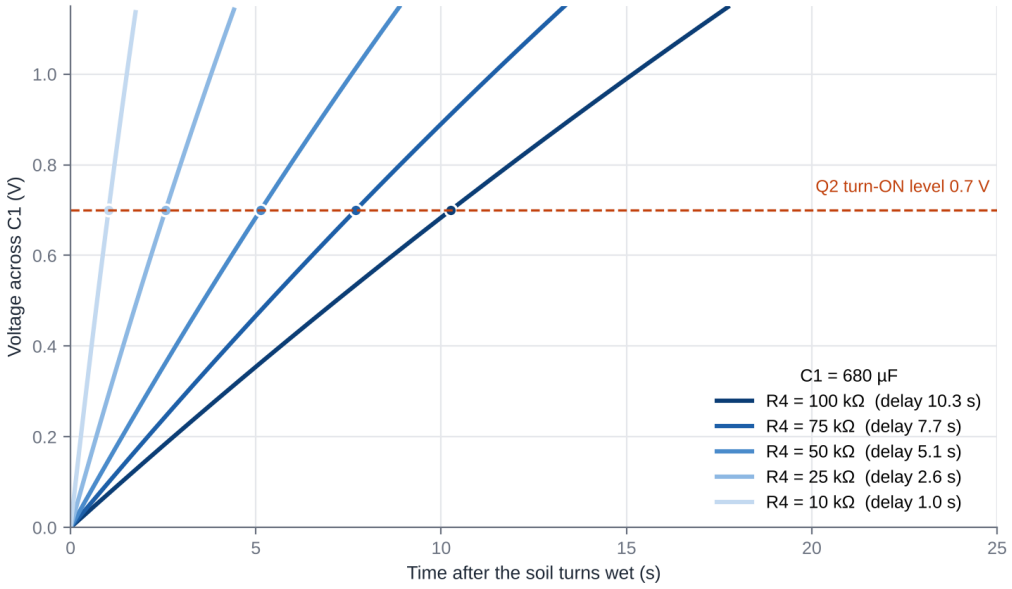


Fig. 6: Charging of C1 for different settings of R4 (C1 = 680 µF)

These are theoretical values. In practice the delay is somewhat longer, because electrolytic capacitors have a tolerance of about ±20 %, the transistor does not turn ON sharply at exactly 0.7 V, and Q3 turns OFF only when Q2 is driven well into conduction. The reference article reports watering times of the order of tens of seconds for the same component values [1]. The delay should therefore be measured on the actual circuit, and C1 can be changed to obtain the time that the plant needs.

3.7.4 Q2 and Q3

When Q2 is ON, the current through R3 is about (5 − 0.2) V / 1 kΩ = 4.8 mA, so Q2 needs a base current of at least 4.8 mA / 110 ≈ 44 µA. At the maximum setting of R4 (100 kΩ), the base current available from the timer is (5 − 0.7) V / 100 kΩ = 43 µA. This is just enough for transistors in the low gain group and ample for the usual BC547B and BC547C parts. At lower settings of R4 the base current is much larger. If the circuit is built with low-gain transistors, R4 should not be set to its extreme maximum.

When Q2 is OFF, R3 supplies the base current of Q3: (5 − 0.7) V / 1 kΩ = 4.3 mA. The pump current that Q3 can carry in saturation depends on this base current and on the transistor rating. Table 7 shows the forced gain required for different pump currents.

Table 7: Suitability of Q3 (BC547) for different pump currents (IB = 4.3 mA)

Pump current	Forced gain needed (IC / IB)	Remark
50 mA	12	Suitable
100 mA	23	Suitable, at the limit of the BC547 current rating
200 mA	47	Exceeds the 100 mA rating, use a stronger transistor
400 mA	93	Exceeds the rating, use a stronger transistor

For a pump that draws more than 100 mA, Q3 should be replaced by a medium-power NPN transistor such as the BC337 or 2N2222 (rated about 800 mA), and R3 can be reduced to about 470 Ω to give more base current. The power lost in Q3 is very small. For example, at 100 mA and VCE(sat) = 0.25 V the dissipation is only 100 mA × 0.25 V = 25 mW, far below the 500 mW rating.

3.7.5 Power and Energy Consumption

The voltage across the pump is about 5 V − 0.2 V = 4.8 V. For a pump that draws 100 mA the power is about 0.5 W while the pump is running. When the pump is OFF, the circuit takes only the current of the sensor module (typically a few milliamperes to about 20 mA because of its comparator and indicator LED) and about 5 mA through R3 when Q2 is ON. A total standby current of about 10 mA to 25 mA is a reasonable estimate. With an average of 20 mA, a 10,000 mAh power bank would run the electronics for about 500 hours, which is about 20 days, and the pump runs only for short periods. These are estimates and should be confirmed with a multimeter.

3.8 Design Considerations

- The pump gives electrical noise and a voltage dip when it starts. A 100 nF ceramic capacitor across the supply, and a short thick supply wire, reduce the chance that the sensor module misbehaves when the pump starts.
- The circuit has no hysteresis of its own. The dryness threshold is set in the sensor module (by its trimmer), and the RC delay prevents rapid switching.
- The delay is a fixed time after the soil becomes wet. The quantity of water delivered therefore depends on the pump flow rate as well as on the delay, and the setting must be adjusted by observation for each plant.
- All emitters must be connected to the same ground as the sensor and the supply. A missing common ground is the most frequent cause of wrong operation.

4. IMPLEMENTATION

4.1 Software Simulation

Before the hardware was assembled, the circuit was built and tested in the TinkerCAD online circuit simulator. The simulation allows the connections to be checked and the behaviour to be observed without risk to the components. The breadboard layout used in the simulation is shown in Fig. 7. A bench power supply provides 5 V to the breadboard rails, the soil moisture sensor is connected to the supply and its signal pin is connected to the transistor circuit, the three transistors and the capacitor are placed on the top half of the breadboard, and the resistors and the potentiometer are on the lower half. The pump is shown at the right-hand side, with its diode connected across it.

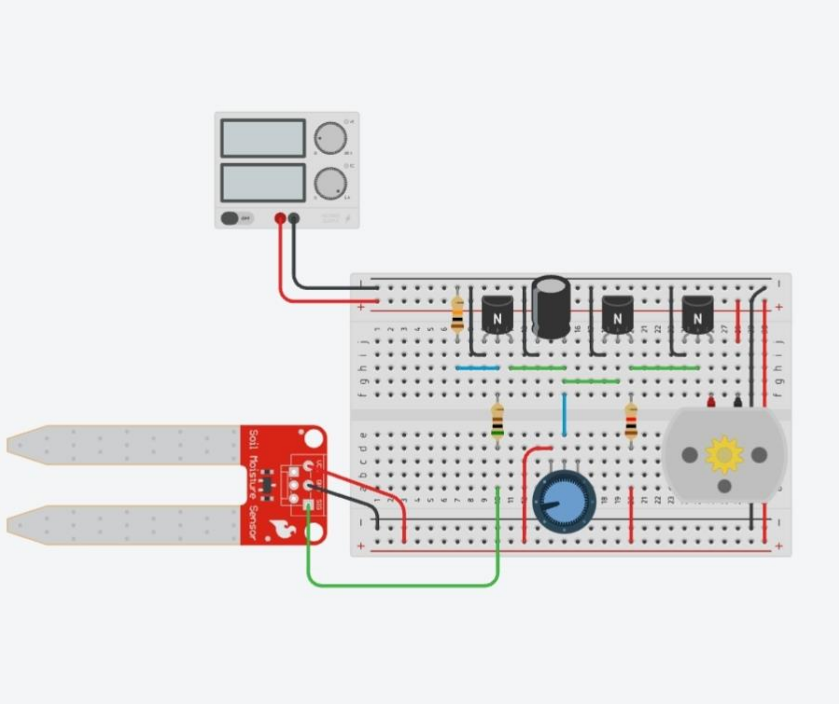


Fig. 7: Breadboard layout of the circuit in the TinkerCAD simulation

The sensor model in the simulator gives an output that is the inverse of the physical module, so the pump behaviour appears reversed in the simulation. Apart from this, the circuit and the logic are the same as in the real hardware.

4.2 Hardware Assembly

4.2.1 Tools and Materials

A breadboard, jumper wires, a small screwdriver for the potentiometer, a wire stripper, a multimeter, a 5 V supply, hot glue or waterproof sealant for the pump wires, a plastic water tube, a bottle of water and a potted plant are required.

4.2.2 Assembly Procedure

1. Mount the three BC547 transistors on the breadboard with the flat face in the same direction, and note the collector, base and emitter positions as shown in Fig. 2.
2. Place the electrolytic capacitor C1 with its negative lead (the lead marked with a stripe) towards the ground rail.
3. Place the resistors R1 (500 Ω), R2 (10 kΩ) and R3 (1 kΩ) and the potentiometer R4 according to the circuit diagram.
4. Connect all the transistor emitters and the negative lead of C1 to the ground rail of the breadboard.
5. Join the collector of Q1, the positive lead of C1, the wiper of R4 and the base of Q2 together to form the timer node. Connect the outer terminal of R4 to +5 V.
6. Connect the collector of Q2 to the base of Q3 and to the free end of R3, and connect the other end of R3 to +5 V.
7. Connect the pump between +5 V and the collector of Q3. Place the diode D1 across the pump terminals, with the cathode (the lead with the band) towards +5 V.
8. Connect the sensor: VCC to +5 V, GND to ground, and SIG to R1. Connect the free end of R1 to the base of Q1 and to R2, and connect the other end of R2 to ground.
9. Before applying power, check all connections against Table 8 and check with the multimeter that there is no short circuit between +5 V and ground.

4.2.3 Connection List

Table 8: Point-to-point connections of the circuit

Component / pin	Connected to	Purpose
Sensor VCC	+5 V rail	Powers the sensor
Sensor GND	Ground rail	Common ground
Sensor SIG	One end of R1	Moisture signal
R1 (other end)	Base of Q1 and R2	Base current limit
R2 (other end)	Ground rail	Pull-down for Q1
Q1 emitter / collector	Ground / timer node	Discharge switch for C1
R4 outer terminal / wiper	+5 V / timer node	Charging resistance
C1 positive / negative	Timer node / ground	Timing capacitor
Q2 base / emitter	Timer node / ground	Inverter input
Q2 collector	R3 and the base of Q3	Inverter output
R3 (other end)	+5 V rail	Pull-up
Q3 emitter / collector	Ground / pump negative	Pump switch
Pump positive	+5 V rail	Pump supply
D1 cathode / anode	+5 V / Q3 collector	Flyback protection

4.3 Power Supply Arrangement

The circuit works from 5 V DC. The options for powering it are listed in Table 9. A power bank is the most convenient choice because it keeps working when the mains supply is off and can be used when travelling.

Table 9: Power supply options

Source	Remarks
USB power bank	Best for portable use and vacation use. A micro-USB breakout board gives access to the +5 V and ground pins.
Mobile phone charger	A 5 V USB wall adapter suits a permanent installation.
5 V DC adapter	A regulated 5 V supply is suitable for continuous use.
Battery with boost converter	A single Li-ion cell with a boost converter and a protection circuit gives a fully wireless version.

4.4 Mechanical Setup

The pump is placed inside a water bottle or container with a wide mouth, so that it is always under water. A plastic tube is connected to the pump outlet and led to the flowerpot. The sensor probes are pushed into the soil, away from the end of the tube

and not too close to the pot wall, so that the sensor reads the moisture of the root zone. The breadboard is kept near the pot but away from the water, and the supply is connected last. The complete arrangement is shown in Fig. 8.

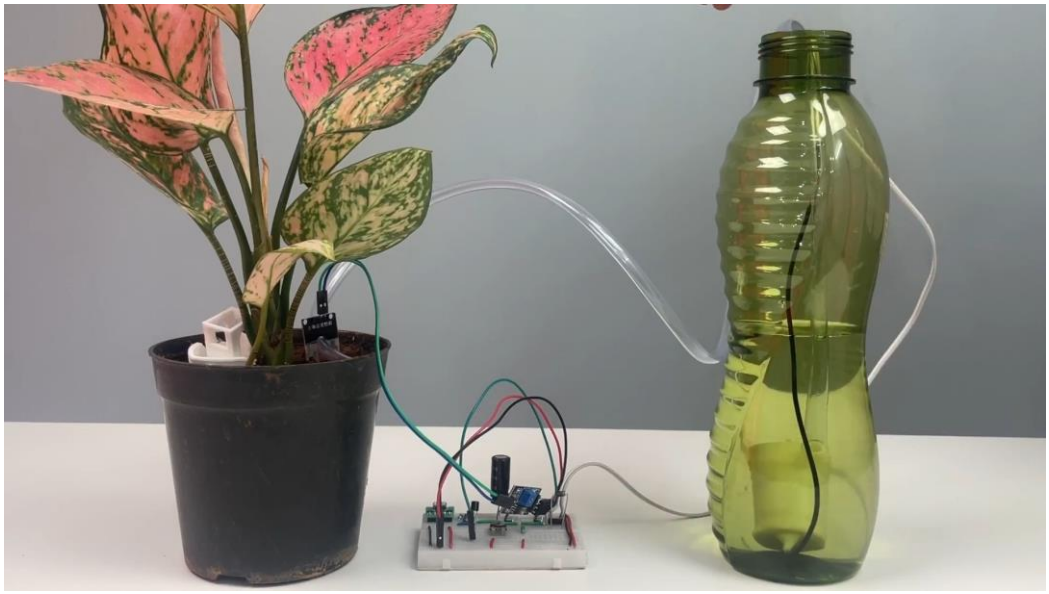


Fig. 8: Final setup: flowerpot with sensor, circuit on the breadboard and the pump in the water bottle

The pump wires were extended and the joints were sealed with hot glue so that no water reaches the copper. The breadboard circuit can later be soldered on a dotted PCB and placed in a small plastic enclosure for a more permanent installation.

4.5 Calibration and Setting the Watering Time

4.5.1 Setting the Dryness Threshold

The moisture level at which the sensor output changes from LOW to HIGH is set by the trimmer on the sensor module. With the probes in the soil of the plant that has just been watered, the trimmer is adjusted until the output is LOW (wet). The soil is then allowed to dry, and the threshold is confirmed by checking that the output turns HIGH when the soil is dry enough to need water.

4.5.2 Setting the Pump Time

1. Turn the potentiometer R4 to its middle position as a starting point.
2. Let the soil dry until the pump starts, then observe how long the pump runs after the soil becomes wet.
3. After about 2 to 4 hours, check the soil. If it is too wet, reduce the resistance of R4 (shorter time). If it is still dry deep in the pot, increase the resistance (longer time).

4. If the range of R4 is not enough, change C1 (a smaller capacitor for a shorter time, a larger capacitor for a longer time) and repeat.

4.6 Testing Procedure

The circuit was tested in stages, from the simplest check to the complete system, as listed in Table 10. Testing in stages makes it easy to find the stage that has a fault.

Table 10: Stage-wise testing procedure and expected results

Step	Test	Expected result
1	Check continuity and look for shorts between +5 V and ground before power-on	No short circuit; all joints correct
2	Apply 5 V and measure the supply at the sensor, the potentiometer and the pump	5 V at all points
3	Keep the sensor probes in dry air	Sensor output HIGH, Q1 ON, pump runs
4	Dip the sensor probes in a glass of water	Sensor output LOW; the pump continues for the delay time and then stops
5	Remove the probes from the water and dry them	Output HIGH again; the pump starts immediately
6	Vary R4 and repeat step 4	Delay changes with the potentiometer setting
7	Insert the probes in the soil of the plant and run the complete system	Pump starts when the soil is dry and stops after the delay when wet

4.7 Precautions

- Keep all electronic parts away from water, and seal the pump leads properly.
- Do not run the pump dry. Keep enough water in the container.
- Connect the electrolytic capacitor and the diode with the correct polarity, and check the transistor pins before power-on.
- Do not turn R4 to zero resistance (see Section 3.7.2).
- Clean the sensor probes regularly and remove them from wet soil when the system is not in use for a long period, since the probes corrode.
- Use a 5 V supply only. A higher voltage can damage the pump and the sensor.
- Do not use the system for plants like cacti that need the soil to remain dry between waterings.

4.8 Cost Estimate

The approximate cost of the components is given in Table 11. The prices are rough values for the Indian market and may change from shop to shop. The total shows that the system is much cheaper than a microcontroller-based one.

Table 11: Approximate cost of the project

Component	Quantity	Approx. cost (₹)
Soil moisture sensor module	1	60 – 100
BC547 transistors	3	5 – 10
Electrolytic capacitor 680 µF	1	5 – 10
Resistors (500 Ω, 1 kΩ, 10 kΩ)	3	3 – 5
Potentiometer 100 kΩ	1	15 – 30
Diode 1N4007	1	1 – 2
Mini submersible pump with tube	1	80 – 150
Breadboard and jumper wires	1 set	100 – 150
Total (approximate)		270 – 450

5. RESULT AND DISCUSSION

5.1 Test Conditions

The system was tested with a potted plant, the sensor probes inserted in the soil, and the pump placed in a bottle of water with the tube leading to the pot. The circuit was powered from a regulated 5 V DC source. The potentiometer was first kept at its middle position and then varied to see the effect on the watering time.

5.2 Observations

The behaviour of the circuit agrees with the design. When the soil is dry, the sensor output is HIGH and the pump starts at once. As the pump supplies water and the soil becomes wet, the sensor output falls to LOW, but the pump does not stop immediately: it continues for the delay set by R4 and C1 and then switches OFF. It stays OFF as long as the soil remains wet. When the soil dries again, the pump starts again. The pump never switches ON and OFF rapidly around the threshold, which was the purpose of the RC timer. The sequence of events is listed in Table 5 and drawn in Fig. 5.

5.3 Voltage Levels at the Nodes

Table 12 lists the voltages expected at the main nodes in the three operating conditions. These values follow from the analysis of the circuit and can be checked with a multimeter during testing; small differences are normal because of transistor and component tolerances.

Table 12: Expected voltages at the main nodes

Node	Soil dry (pump ON)	Soil wet, delay running (pump ON)	Soil wet, delay over (pump OFF)
Sensor SIG output	about 5 V (HIGH)	about 0 V (LOW)	about 0 V (LOW)
Base of Q1	about 0.7 V	about 0 V	about 0 V
Timer node (across C1)	about 0.1 – 0.2 V	rising from 0.2 V to 0.7 V	about 0.7 V
Collector of Q2 / base of Q3	about 0.7 V	about 0.7 V	about 0.1 – 0.2 V
Collector of Q3	about 0.1 – 0.25 V	about 0.1 – 0.25 V	about 5 V
Voltage across the pump	about 4.8 V	about 4.8 V	0 V

5.4 Effect of the Potentiometer and Capacitor

The pump-ON delay is directly proportional to the resistance of R4 and the capacitance of C1, as shown by the calculated values in Table 6 and by the straight lines in Fig. 9. Turning R4 to a higher resistance increases the watering time and turning it to a lower resistance decreases it. For plants with different needs, the setting can be chosen as suggested in Table 13.

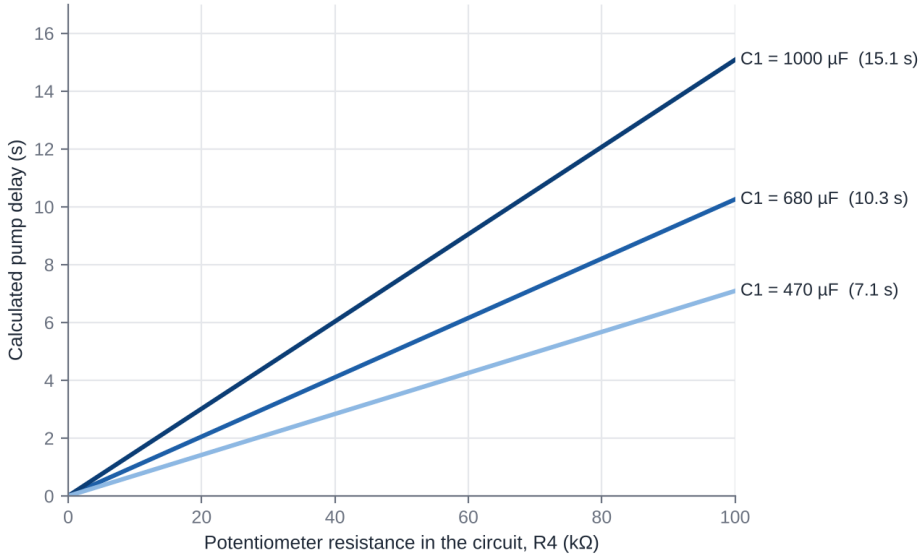


Fig. 9: Calculated pump delay against the resistance of R4 for three values of C1

Table 13: Suggested setting for different types of plants

Type of plant	Water requirement	Suggested setting
Succulents and similar low-water plants	Low	Low R4 (short delay), or a smaller C1
Pothos, spider plant and similar	Moderate	Middle position of R4
Ferns, peace lily and similar	High	High R4 (long delay), or a larger C1
Cactus and desert plants	Soil must stay dry	Automatic watering is not suitable

5.5 Discussion

5.5.1 Role of Each Transistor

Each transistor has a clear function. Q1 is a discharge switch that resets the timer whenever the soil is dry. Q2 inverts the timer voltage, and Q3 handles the pump current. Q2 could drive the pump directly if the sensor gave a LOW output for dry soil, but since most sensors give HIGH for dry soil, an extra inversion is needed. Separating

the tasks of timing and driving also keeps the small timing current away from the heavy pump current.

5.5.2 Importance of the RC Delay

Without the RC timer, the pump would stop as soon as the sensor output went LOW. The probes touch only a small part of the soil, so the sensor would read wet within a very short time and the pump would run for a fraction of a second, giving too little water. As the water spreads and the reading changes, the pump would also start and stop repeatedly. The RC timer keeps the pump running for a set time after wetting, so that the soil receives enough water in a single cycle.

5.5.3 Pump Current and Transistor Rating

A BC547 is rated for 100 mA, while small pumps can draw 200 mA to 400 mA. If the pump draws more than the rating, Q3 should be replaced as discussed in Section 3.7.4. A small pump of low current can be driven directly by the BC547.

5.5.4 Overwatering Protection

The sensor prevents watering when the soil is already wet, since the pump starts only when the soil is dry. However, the timer sets only the duration of the watering and does not measure the quantity of water given. The setting of R4 and C1 must therefore be adjusted to the plant, and the water outlet should be kept well away from the sensor probes.

5.6 Advantages

- Very low cost, with a total of a few hundred rupees.
- No microcontroller or programming is needed.
- Uses common components that are easily available.
- The watering time is adjustable with a single potentiometer.
- Protection of the driver transistor by the flyback diode.
- Low power consumption, and works from a power bank or an adapter.
- Simple to troubleshoot by measuring node voltages.

5.7 Limitations

- The BC547 can switch only small pumps (up to about 100 mA) without a change of Q3.
- The resistive sensor probes corrode with time.

- The delay depends on the tolerance of the electrolytic capacitor and on the transistor gain, and it can change with temperature and ageing.
- The system does not detect an empty water tank, so the pump may run dry.
- There is no display, indication or remote alert.
- The system cannot water on a schedule and cannot change its behaviour without changing the hardware.

5.8 Troubleshooting

Table 14: Common problems, causes and solutions

Problem	Possible cause	Solution
Pump runs continuously	Sensor output inverted, Q2 faulty or no common ground	Check the sensor output polarity, test Q2 and check the ground connections
Pump never starts	No supply, wrong transistor orientation or faulty Q3	Check 5 V at all points, the transistor pins and the diode direction
Watering time too short	C1 too small or R4 at a low value	Increase C1 or turn R4 to a higher resistance
Watering time too long	C1 too large or R4 at maximum	Reduce C1 or lower the resistance of R4
Sensor not detecting moisture	Corroded probes or poor contact with the soil	Clean the probes and insert them properly
Weak water flow	Low voltage at the pump or kinked tube	Check 5 V at the pump and straighten the tube
Pump or sensor restarts when the pump starts	Supply voltage dips	Use a stronger supply and add a 100 nF capacitor across the supply

5.9 Comparison with Microcontroller-Based Systems

Table 15: Comparison between the proposed system and an Arduino-based system

Feature	Proposed transistor-based system	Arduino-based system
Programming	None	Required
Components	Few discrete components	Board, driver or relay module and more wiring
Cost	Low	Higher
Features	Single sensor, adjustable watering time	Display, data logging, timed watering, remote access
Changing behaviour	Needs a hardware change	Change in the program
Debugging	Measure node voltages with a multimeter	Serial monitor and code debugging

The microcontroller-based system is more flexible, but for the watering of a potted plant the proposed system gives the necessary function at a lower cost and with less effort.

5.10 Conclusion

An automatic plant watering system was designed, simulated, assembled and tested without using a microcontroller. The soil moisture sensor, three BC547 transistors and an RC timer together switch a small DC pump: the pump runs when the soil is dry and stops after an adjustable delay once the soil is wet. The diode across the pump protects the driver transistor, and the whole circuit works from a single 5 V supply. The design calculations show that all the transistors are driven into proper saturation, and that the watering time is proportional to the product of R4 and C1.

The project met its objectives. It is simple, inexpensive and reliable for small potted plants, and it gives a practical application of transistor switching, an inverter and an RC time delay that are studied in the Analog and Digital Electronics laboratory.

5.11 Future Scope

- Replace Q3 by a stronger transistor, or use a relay or a MOSFET, to drive larger pumps or a solenoid valve.
- Use a capacitive soil moisture sensor to avoid probe corrosion.
- Add a water-level sensor with an LED or buzzer to warn when the tank is empty and to stop the pump from running dry.
- Add a comparator with hysteresis (Schmitt trigger) for a cleaner switching action.
- Power the system from a solar panel and a battery for outdoor use.
- Use a manifold and one pump, or separate circuits, to water several plants.
- Convert the breadboard circuit into a soldered PCB in a waterproof enclosure.
- Add a microcontroller or an IoT module later for monitoring and remote control.

6. REFERENCES

[1] R. Krisna, "A Simple Automatic Plant Watering System without Microcontroller," CircuitDigest, May 2024. https://circuitdigest.com/electronic-circuits/automatic-plant-watering-system-without-arduino

[2] BC547 NPN transistor – pinout and datasheet, Components101. https://components101.com/transistors/bc547-transistor-pinout-datasheet

[3] Soil moisture sensor module, Components101. https://components101.com/modules/soil-moisture-sensor-module

[4] BC546/BC547/BC548/BC549 NPN epitaxial silicon transistor datasheet, Fairchild Semiconductor / onsemi.

[5] 1N4001–1N4007 general-purpose rectifier diode datasheet, onsemi.

[6] A. S. Sedra and K. C. Smith, Microelectronic Circuits, Oxford University Press.

[7] R. L. Boylestad and L. Nashelsky, Electronic Devices and Circuit Theory, Pearson.

[8] TinkerCAD Circuits, Autodesk. https://www.tinkercad.com